

Find all solutions to each equation in degrees.

9) $\sin 4x + \sin 6x = 0$

10) $0 = \cos 2x + \cos 6x$

11) $0 = \cos 8x + \cos 6x$

12) $\cos x + \cos 7x = 0$

13) $\sin 8x + \sin 2x = \cos 3x$

14) $\sin 3x + \sin 2x + \sin 4x = 0$

15) $\sin 2x = -\cos 3x + \sin 4x$

16) $0 = \sin x + \sin 3x + \sin 5x$

Critical thinking question:

17) Write an equation involving $\cos x$ and $\sin \frac{x}{2}$ that has no solution.